# Midterm Review: Chapters 1–3

Every theorem through one matrix (and its twin) — full calculations

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# One matrix carries the midterm

## The cast — one matrix and its twin

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 5 & 3 \\ 1 & 0 & 8 \end{bmatrix} \qquad A' = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 5 & 3 \\ 3 & 7 & 6 \end{bmatrix} \quad (\text{row 3 } \textit{replaced} \text{ by } R_1 + R_2)$$

$A$: $\det = -1$, invertible. $A'$: $\det = 0$, singular — one replaced row turns the whole theory around. Reused from the tutorial: for $\mathbf{b} = (-3, 1, -15)$, $A\mathbf{x} = \mathbf{b}$ has $\mathbf{x} = (1, 1, -2)$.

| Chapter | Theorem themes | How the pair shows them |
|---|---|---|
| Ch. 1 | elimination, existence & uniqueness, spans, independence, transformations | $A$ solves uniquely (slide 4); $A'$ shows none / infinitely many (slides 5, 6) |
| Ch. 2 | matrix algebra, inverses, the Invertible Matrix Theorem, subspaces, rank | every IMT statement true for $A$, false for $A'$ (slide 12) |
| Ch. 3 | cofactor expansion, properties, Cramer, adjugate, volume | det as the one-number certificate: $-1$ vs. 0 (slides 15–19) |

# Ch. 1 theorem inventory (1.1–1.4)

| Theorem | Statement (sketch) | Worked where |
|---|---|---|
| Existence & Uniqueness (1.2) | consistent $\iff$ rightmost column of $[A \mid \mathbf{b}]$ is not a pivot column; unique $\iff$ consistent and no free variable | slides 4, 5 |
| Uniqueness of RREF (1.2) | each matrix has exactly one reduced echelon form | every elimination here |
| Row equivalence (1.1–1.2) | row operations are reversible; row-equivalent systems have the same solution set | slide 4 |
| Vector equation (1.3) | $x_1\mathbf{a}_1 + \cdots + x_n\mathbf{a}_n = \mathbf{b}$ and $A\mathbf{x} = \mathbf{b}$ share solution sets; $\mathbf{b} \in \mathrm{Span}\{\mathbf{a}_i\}$ $\iff$ consistent | slide 7 |
| Spanning test (Thm 1.4) | $A\mathbf{x} = \mathbf{b}$ consistent for *every* $\mathbf{b}$ $\iff$ columns span $\mathbb{R}^m$ | slide 7 |

# Ch. 1 step by step: solve $A\mathbf{x} = \mathbf{b}$, $\mathbf{b} = (-3, 1, -15)$

**Forward pass** (add multiples of a row — solution set unchanged):

$$\left[\begin{array}{ccc|c} 1 & 2 & 3 & -3 \\ 2 & 5 & 3 & 1 \\ 1 & 0 & 8 & -15 \end{array}\right] \xrightarrow[R_3 \to R_3 - R_1]{R_2 \to R_2 - 2R_1} \left[\begin{array}{ccc|c} 1 & 2 & 3 & -3 \\ 0 & 1 & -3 & 7 \\ 0 & -2 & 5 & -12 \end{array}\right] \xrightarrow{R_3 \to R_3 + 2R_2} \left[\begin{array}{ccc|c} 1 & 2 & 3 & -3 \\ 0 & 1 & -3 & 7 \\ 0 & 0 & -1 & 2 \end{array}\right]$$

**Backward pass** (scale the pivot, clear above):

$$\left[\begin{array}{ccc|c} 1 & 2 & 3 & -3 \\ 0 & 1 & -3 & 7 \\ 0 & 0 & -1 & 2 \end{array}\right] \xrightarrow{R_3 \to -R_3} \left[\begin{array}{ccc|c} 1 & 2 & 3 & -3 \\ 0 & 1 & -3 & 7 \\ 0 & 0 & 1 & -2 \end{array}\right] \xrightarrow[R_2 \to R_2 + 3R_3]{R_1 \to R_1 - 3R_3} \left[\begin{array}{ccc|c} 1 & 2 & 0 & 3 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & -2 \end{array}\right]$$

$$\xrightarrow{R_1 \to R_1 - 2R_2} \left[\begin{array}{ccc|c} 1 & 0 & 0 & 1 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & -2 \end{array}\right] \qquad \text{RREF} \qquad \mathbf{x} = (1, 1, -2)$$

## Read off — and tag the theorems

Every variable column has a pivot and the rightmost column does not: **consistent** (Existence & Uniqueness Thm) with **no free variable**: **unique**. $\mathbf{x} = (1, 1, -2)$. Row equivalence (each op reversible) means we may substitute back: $1 + 2 - 6 = -3$, $2 + 5 - 6 = 1$, $1 + 0 - 16 = -15$ ✓

# Ch. 1 step by step: the twin with $\mathbf{b} = (1, 0, 5)$ — no solution

**Same forward pass** on $[\,A' \mid \mathbf{b}\,]$:

$$\left[\begin{array}{ccc|c} 1 & 2 & 3 & 1 \\ 2 & 5 & 3 & 0 \\ 3 & 7 & 6 & 5 \end{array}\right] \xrightarrow[R_3 \to R_3 - 3R_1]{R_2 \to R_2 - 2R_1} \left[\begin{array}{ccc|c} 1 & 2 & 3 & 1 \\ 0 & 1 & -3 & -2 \\ 0 & 1 & -3 & 2 \end{array}\right] \xrightarrow{R_3 \to R_3 - R_2} \left[\begin{array}{ccc|c} 1 & 2 & 3 & 1 \\ 0 & 1 & -3 & -2 \\ 0 & 0 & 0 & 4 \end{array}\right]$$

$$\xrightarrow{R_1 \to R_1 - 2R_2} \left[\begin{array}{ccc|c} 1 & 0 & 9 & 5 \\ 0 & 1 & -3 & -2 \\ 0 & 0 & 0 & 4 \end{array}\right]$$

## The forbidden row (Existence & Uniqueness Thm)

Last row: $0\,x_1 + 0\,x_2 + 0\,x_3 = 4$ — impossible. The rightmost column *is* a pivot column $\Rightarrow$ **inconsistent**, no solution. Why it had to fail: row 3 of $A'$ equals row 1 + row 2, so any solution would need $b_3 = b_1 + b_2$; here $5 \neq 1 + 0$.

- Compare $A$ (previous slide): identical machinery, opposite verdict — only the right-hand side decides.

# Ch. 1 step by step: the twin with $\mathbf{b} = (1, 1, 2)$ — infinitely many

**Forward pass**: the same eliminations now end in $0 = 0$:

$$\left[\begin{array}{ccc|c} 1 & 2 & 3 & 1 \\ 2 & 5 & 3 & 1 \\ 3 & 7 & 6 & 2 \end{array}\right] \xrightarrow[R_3 \to R_3 - 3R_1]{R_2 \to R_2 - 2R_1} \left[\begin{array}{ccc|c} 1 & 2 & 3 & 1 \\ 0 & 1 & -3 & -1 \\ 0 & 1 & -3 & -1 \end{array}\right] \xrightarrow{R_3 \to R_3 - R_2} \left[\begin{array}{ccc|c} 1 & 2 & 3 & 1 \\ 0 & 1 & -3 & -1 \\ 0 & 0 & 0 & 0 \end{array}\right]$$

$$\xrightarrow{R_1 \to R_1 - 2R_2} \left[\begin{array}{ccc|c} 1 & 0 & 9 & 3 \\ 0 & 1 & -3 & -1 \\ 0 & 0 & 0 & 0 \end{array}\right]$$

**Read off** with free variable $x_3$: $x_1 = 3 - 9x_3$, $x_2 = -1 + 3x_3$, $x_3$ free. Split into particular + homogeneous (Thm 1.5):

$$\mathbf{x} = \begin{bmatrix} 3 \\ -1 \\ 0 \end{bmatrix} + x_3 \begin{bmatrix} -9 \\ 3 \\ 1 \end{bmatrix} \qquad A' \begin{bmatrix} 3 \\ -1 \\ 0 \end{bmatrix} = (1, 1, 2)\ \checkmark \qquad A' \begin{bmatrix} -9 \\ 3 \\ 1 \end{bmatrix} = (-9 + 6 + 3,\ -18 + 15 + 3,\ -27 + 21 + 6) = \mathbf{0}\ \checkmark$$

- One free variable $\Rightarrow$ a *line* of solutions; the direction is a nontrivial solution of $A'\mathbf{x} = \mathbf{0}$ — the twin's columns are dependent.

# Ch. 1 step by step: vectors, spans, independence

## Vector equation: the weights are the solution

$\mathbf{x} = (1, 1, -2)$ from slide 4 means

$$\mathbf{b} = 1\,\mathbf{a}_1 + 1\,\mathbf{a}_2 - 2\,\mathbf{a}_3$$

Check:
$(1 + 2 - 6,\ 2 + 5 - 6,\ 1 + 0 - 16) = (-3, 1, -15)$ ✓
— the system and the vector equation are the same question (Thm 1.3), and $\mathbf{b} \in \mathrm{Span}\{\mathbf{a}_1, \mathbf{a}_2, \mathbf{a}_3\}$.

## Linearity demo (1.8)

$T(\mathbf{x}) = A\mathbf{x}$ with $\mathbf{e}_1 = (1, 0, 0)$:
$T(2\mathbf{e}_1) = A(2, 0, 0) = (2, 4, 2)$
and $2\,T(\mathbf{e}_1) = 2(1, 2, 1) = (2, 4, 2)$ ✓

## Independence: solve $A\mathbf{x} = \mathbf{0}$ (Thm 1.7)

$$\left[\begin{array}{ccc|c} 1 & 2 & 3 & 0 \\ 2 & 5 & 3 & 0 \\ 1 & 0 & 8 & 0 \end{array}\right] \xrightarrow[R_3 \to R_3 - R_1]{R_2 \to R_2 - 2R_1} \left[\begin{array}{ccc|c} 1 & 2 & 3 & 0 \\ 0 & 1 & -3 & 0 \\ 0 & -2 & 5 & 0 \end{array}\right] \xrightarrow{R_3 \to R_3 + 2R_2} \left[\begin{array}{ccc|c} 1 & 2 & 3 & 0 \\ 0 & 1 & -3 & 0 \\ 0 & 0 & -1 & 0 \end{array}\right]$$

Back-substitute:
$x_3 = 0 \Rightarrow x_2 = 3x_3 = 0 \Rightarrow x_1 = -2x_2 - 3x_3 = 0$. Only the trivial solution $\Rightarrow$ columns **independent** $\Rightarrow$ $T$ **one-to-one** (Thm 1.9); square with rank 3 $\Rightarrow$ also **onto**. For $A'$: $(-9, 3, 1)$ is annihilated $\Rightarrow$ dependent, neither.

# Ch. 1 theorem inventory (1.5–1.9)

| Theorem | Statement (sketch) | Worked where |
|---|---|---|
| Homogeneous systems (1.5) | $A\mathbf{x} = \mathbf{0}$ has a nontrivial solution $\iff$ free variable $\iff$ columns dependent | slide 6 |
| Particular + homogeneous (Thm 1.5) | if $A\mathbf{x} = \mathbf{b}$ is consistent, *all* solutions are $\mathbf{x}_p + \mathbf{v}_h$ | slide 6 |
| Independence tests (1.7) | $\geq 2$ vectors: dependent $\iff$ one is a combination of the others; more vectors than entries $\Rightarrow$ dependent | slide 7 |
| Linear transformations (1.8) | $T$ linear $\iff$ $T(\mathbf{u}+\mathbf{v}) = T\mathbf{u} + T\mathbf{v}$ and $T(c\mathbf{u}) = c\,T\mathbf{u}$; every $A$ gives linear $\mathbf{x} \mapsto A\mathbf{x}$ | slide 7 |
| Onto / one-to-one (Thm 1.9) | $T$ onto $\iff$ columns span $\mathbb{R}^m$; one-to-one $\iff$ columns linearly independent (equivalently $A\mathbf{x} = \mathbf{0}$ only trivial) | slide 7 |

# Ch. 2 theorem inventory (2.1–2.5)

| Theorem | Statement (sketch) | Worked where |
|---|---|---|
| Algebra laws (2.1) | $(AB)C = A(BC)$, distributivity; $AB \neq BA$ in general; $(AB)^\top = B^\top A^\top$ | slide 11 |
| Inverses (2.2) | inverse unique; $2 \times 2$: $A^{-1} = \frac{1}{ad-bc}\begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$; $(AB)^{-1} = B^{-1}A^{-1}$; $(A^\top)^{-1} = (A^{-1})^\top$ | slides 10, 11 |
| Elementary matrices (2.2) | one row op on $I$; multiplying by $E$ performs that op; invertible $\iff$ row equivalent to $I$; $[A \mid I] \sim [I \mid A^{-1}]$ | slide 10 |
| Column–row expansion (2.4) | $AB = \sum_j \mathrm{col}_j(A)\,\mathrm{row}_j(B)$ | slide 11 |
| LU factorization (2.5) | row replacements only $\Rightarrow A = LU$, $L$ unit lower triangular | $A$ reduces with additions only: $LU$ exists |

# Ch. 2 step by step: $A^{-1}$ via $[A \mid I] \sim [I \mid A^{-1}]$

**Forward pass** (the same three ops as always), two steps:

$$\left[\begin{array}{ccc|ccc} 1 & 2 & 3 & 1 & 0 & 0 \\ 2 & 5 & 3 & 0 & 1 & 0 \\ 1 & 0 & 8 & 0 & 0 & 1 \end{array}\right] \xrightarrow[R_3 \to R_3 - R_1]{R_2 \to R_2 - 2R_1} \left[\begin{array}{ccc|ccc} 1 & 2 & 3 & 1 & 0 & 0 \\ 0 & 1 & -3 & -2 & 1 & 0 \\ 0 & -2 & 5 & -1 & 0 & 1 \end{array}\right] \xrightarrow{R_3 \to R_3 + 2R_2} \left[\begin{array}{ccc|ccc} 1 & 2 & 3 & 1 & 0 & 0 \\ 0 & 1 & -3 & -2 & 1 & 0 \\ 0 & 0 & -1 & -5 & 2 & 1 \end{array}\right]$$

**Backward pass** (scale the pivot, then clear upward), three steps:

$$\xrightarrow{R_3 \to -R_3} \left[\begin{array}{ccc|ccc} 1 & 2 & 3 & 1 & 0 & 0 \\ 0 & 1 & -3 & -2 & 1 & 0 \\ 0 & 0 & 1 & 5 & -2 & -1 \end{array}\right] \xrightarrow[R_2 \to R_2 + 3R_3]{R_1 \to R_1 - 3R_3} \left[\begin{array}{ccc|ccc} 1 & 2 & 0 & -14 & 6 & 3 \\ 0 & 1 & 0 & 13 & -5 & -3 \\ 0 & 0 & 1 & 5 & -2 & -1 \end{array}\right]$$

$$\xrightarrow{R_1 \to R_1 - 2R_2} \left[\begin{array}{ccc|ccc} 1 & 0 & 0 & -40 & 16 & 9 \\ 0 & 1 & 0 & 13 & -5 & -3 \\ 0 & 0 & 1 & 5 & -2 & -1 \end{array}\right]$$

## Theorem 7 delivered

Reached $[I \mid A^{-1}]$: $A$ is row equivalent to $I \Rightarrow$ invertible, with $A^{-1} = \left[\begin{smallmatrix} -40 & 16 & 9 \\ 13 & -5 & -3 \\ 5 & -2 & -1 \end{smallmatrix}\right]$. Check $AA^{-1} = I$: row 1 $\cdot$ col 1: $1(-40) + 2(13) + 3(5) = 1$ ✓; row 1 $\cdot$ col 2: $1(16) + 2(-5) + 3(-2) = 0$ ✓

# Ch. 2 step by step: algebra rules with $B = \mathrm{diag}(2, 1, 3)$

## $AB$: columns scaled (column–row expansion, Thm 2.4)

$\mathrm{col}_j(AB) = d_j\,\mathrm{col}_j(A)$, i.e. $AB = \begin{bmatrix} 2 & 2 & 9 \\ 4 & 5 & 9 \\ 2 & 0 & 24 \end{bmatrix}$ with

$$\mathrm{col}_1(AB) = 2\,\mathrm{col}_1(A) = (2, 4, 2)$$
$$\mathrm{col}_2(AB) = \mathrm{col}_2(A) = (2, 5, 0)$$
$$\mathrm{col}_3(AB) = 3\,\mathrm{col}_3(A) = (9, 9, 24)$$

## $BA$: rows scaled — hence $AB \neq BA$

$\mathrm{row}_i(BA) = d_i\,\mathrm{row}_i(A)$:

$$BA = \begin{bmatrix} 2 & 4 & 6 \\ 2 & 5 & 3 \\ 3 & 0 & 24 \end{bmatrix} \neq AB$$

## $(AB)^{-1} = B^{-1}A^{-1}$, proved by associativity

$$(AB)(B^{-1}A^{-1}) = A\,(BB^{-1})\,A^{-1} = A\,I\,A^{-1} = I$$

The reverse product works the same way, so the order *must* reverse. Numerically: $B^{-1} = \mathrm{diag}(\frac{1}{2}, 1, \frac{1}{3})$ and $(AB)^{-1} = B^{-1}A^{-1}$ ✓. ($A'$: rank $2 < 3$ never reaches $I$ — no inverse, $\det A' = 0$, slide 16.)

## Transpose rule spot check

$(AB)^\top = B^\top A^\top$: entry $(1, 1)$ of $(AB)^\top$ is 2; and $\mathrm{row}_1(B^\top) \cdot \mathrm{col}_1(A^\top) = (2, 0, 0) \cdot (1, 2, 1) = 2$ ✓

# The Invertible Matrix Theorem: one matrix, one web

| Statement (all equivalent for $n \times n$ $A$) | $A$ | $A'$ | test used |
|---|---|---|---|
| (a) $A$ invertible | ✓ | × | $\det \neq 0$ (fastest) |
| (b) row equivalent to $I$; (c) $n$ pivot positions | ✓ | × | rank $= 3$ vs. 2 |
| (d) $A\mathbf{x} = \mathbf{0}$ only trivial; (e) columns independent | ✓ | × | $A'(-9, 3, 1) = \mathbf{0}$ |
| (f) $T$ one-to-one | ✓ | × | Ch. 1 Thm 1.9 |
| (g) $A\mathbf{x} = \mathbf{b}$ consistent for every $\mathbf{b}$; (h) columns span; (i) $T$ onto | ✓ | × | $A'$: $b_3 \neq b_1 + b_2$ fails |
| (j), (k) $CA = I$ exists; (l) $A^\top$ invertible | ✓ | × | $A^{-1}$ computed (slide 10); $\det A^\top = -1$ |
| (m) columns form a basis of $\mathbb{R}^n$; (n) $\operatorname{Col} A = \mathbb{R}^n$ | ✓ | × | slide 13 |
| (o) $\operatorname{rank} A = n$; (p) $\dim \operatorname{Nul} A = 0$; (q) $\operatorname{Nul} A = \{\mathbf{0}\}$ | ✓ | × | rank theorem |

- One false statement drags them all down — and $\det A' = 0$ is the cheapest way to see it. For $A$, every row is a ✓: that is what "invertible" means.

# Subspaces and the Rank Theorem (2.8–2.9), step by step

## Definitions

Subspace: contains $\mathbf{0}$, closed under $+$ and scalar mult. $\operatorname{Nul} A = \{\text{solutions of } A\mathbf{x} = \mathbf{0}\}$ and $\operatorname{Col} A = \operatorname{Span}\{\text{columns}\}$ are subspaces. Basis: independent spanning set. **Thm 13**: the pivot columns of the *original* $A$ (not of the RREF!) form a basis for $\operatorname{Col} A$. **Rank Theorem**: $\operatorname{rank}(A) + \dim \operatorname{Nul} A = n$.

## $A$: full rank

$\operatorname{Nul} A = \{\mathbf{0}\}$ (computed on slide 7): basis empty, $\dim 0$. $\operatorname{Col} A = \mathbb{R}^3$ (rank 3): basis = all three columns. Rank Theorem: $3 + 0 = 3$ ✓

## $A'$: the trade-off

$\operatorname{Nul} A' = \operatorname{span}\{(-9, 3, 1)\}$ from the parametric solution (slide 6): $\dim 1$. $\operatorname{Col} A'$: pivot columns 1, 2 are the basis — a plane. The dependency from the null vector $-9\mathbf{c}_1 + 3\mathbf{c}_2 + \mathbf{c}_3 = \mathbf{0}$ reads $\mathbf{c}_3 = 9\mathbf{c}_1 - 3\mathbf{c}_2 = 9(1, 2, 3) - 3(2, 5, 7) = (3, 3, 6)$ ✓ Rank Theorem: $2 + 1 = 3$ ✓

- Free variables live in Nul, pivots in Col; the two dimensions always add to $n = 3$ — the twin shows both extremes of the trade-off.

# Ch. 3 theorem inventory (3.1–3.3)

| Theorem | Statement (sketch) | Worked where |
|---|---|---|
| Cofactor expansion (3.1) | $\det A = \sum_j a_{ij}(-1)^{i+j} M_{ij}$, any row or column | slide 15 |
| Triangular matrices (Thm 3.1) | det = product of diagonal entries | slide 16 |
| Row operations (3.2) | add a multiple: free; swap: sign flips; scale by $k$: det times $k$ | slide 16 |
| Invertibility (3.2) | $A$ invertible $\iff \det A \neq 0$ | slides 15–16 |
| Transpose & product (3.2) | $\det A^\top = \det A$; $\det(AB) = \det A \det B$ | slide 17 |
| Cramer's rule (3.3) | $x_i = \det A_i(\mathbf{b}) / \det A$ (invertible $A$) | slide 18 |
| Adjugate (3.3) | $\operatorname{adj}(A)\, A = (\det A) I$, so $A^{-1} = \operatorname{adj}(A) / \det A$ | slide 19 |
| Area / volume (3.3) | $\lvert\det A\rvert$ = area ($2 \times 2$) or volume ($3 \times 3$) of the column box | slide 16 |

# Ch. 3 step by step: cofactor expansion of det $A$ along row 1

Minors: delete row 1 and the entry's column, take the $2 \times 2$ det:

$$M_{11} = \det \begin{bmatrix} 5 & 3 \\ 0 & 8 \end{bmatrix} = 5{\cdot}8 - 3{\cdot}0 = 40, \quad M_{12} = \det \begin{bmatrix} 2 & 3 \\ 1 & 8 \end{bmatrix} = 16 - 3 = 13, \quad M_{13} = \det \begin{bmatrix} 2 & 5 \\ 1 & 0 \end{bmatrix} = 0 - 5 = -5$$

Signs alternate $+, -, +$ along row 1 (that is $(-1)^{1+j}$):

$$\det A = 1 \cdot (+40) \; - \; 2 \cdot (13) \; + \; 3 \cdot (-5) = 40 - 26 - 15 = \boxed{-1}$$

**Cross-check along column 2** (the zero kills one minor): entries $2, 5, 0$, signs $-, +, -$, minors 13, 5, $-3$:

$$\det A = -2(13) + 5(5) - 0(-3) = -26 + 25 = \boxed{-1} \; \checkmark$$

- Same value from any row or column — and $-1 \neq 0$ already certifies: invertible, rank 3, unique solutions everywhere.

# Ch. 3 step by step: the twin — two routes to $\det A' = 0$

**Route 1 — cofactor expansion along row 1:**

$$M_{11} = \det \begin{bmatrix} 5 & 3 \\ 7 & 6 \end{bmatrix} = 30 - 21 = 9, \quad M_{12} = \det \begin{bmatrix} 2 & 3 \\ 3 & 6 \end{bmatrix} = 12 - 9 = 3, \quad M_{13} = \det \begin{bmatrix} 2 & 5 \\ 3 & 7 \end{bmatrix} = 14 - 15 = -1$$

$$\det A' = 1(9) - 2(3) + 3(-1) = 9 - 6 - 3 = \boxed{0}$$

**Route 2 — row reduce (additions only: det unchanged):**

$$\begin{bmatrix} 1 & 2 & 3 \\ 2 & 5 & 3 \\ 3 & 7 & 6 \end{bmatrix} \xrightarrow[R_3 \to R_3 - 3R_1]{R_2 \to R_2 - 2R_1} \begin{bmatrix} 1 & 2 & 3 \\ 0 & 1 & -3 \\ 0 & 1 & -3 \end{bmatrix} \xrightarrow{R_3 \to R_3 - R_2} \begin{bmatrix} 1 & 2 & 3 \\ 0 & 1 & -3 \\ 0 & 0 & 0 \end{bmatrix} \qquad \det = 1 \cdot 1 \cdot 0 = 0 \ \checkmark$$

## Handle with care

$A'$ was built by *replacing* row 3 with $R_1 + R_2$ — that is *not* one of the three elementary operations, so det was allowed to jump from $-1$ to 0. Only swap / scale / add-a-multiple carry the guaranteed effects listed in the inventory.

- Volume reading: $|\det A| = 1$ — the column parallelepiped fills one unit of volume; $|\det A'| = 0$ — its columns lie flat in a plane.

# Ch. 3 step by step: transpose and product rules

## $\det A^\top = \det A$

$A^\top = \begin{bmatrix} 1 & 2 & 1 \\ 2 & 5 & 0 \\ 3 & 3 & 8 \end{bmatrix}$; expand along row 1, minors 40, 16, $-9$:

$$\det A^\top = 1(40) - 2(16) + 1(-9) = -1 = \det A \ \checkmark$$

## $\det B$ for $B = \mathrm{diag}(2, 1, 3)$ (Thm 3.1)

Diagonal is triangular: $\det B = 2 \cdot 1 \cdot 3 = 6$.

## $\det(AB) = \det A \cdot \det B$ — twice

Product rule: $\det(AB) = (-1)(6) = -6$.
Column-scaling rule: $AB$ has column 1 scaled by 2 and column 3 by 3 (slide 11), so
$\det(AB) = 2 \cdot 3 \cdot \det A = 6(-1) = -6$ ✓
Direct expansion of $AB$ along column 2:

$$-2(96 - 18) + 5(48 - 18) = -156 + 150 = -6$$

- $\det A^\top = \det A$ is *why* every row rule has a column twin — the glue between the two checks above.
- Caution: $\det(A + B)$ has *no* such rule.

# Ch. 3 step by step: Cramer's rule, three determinants

Solve $A\mathbf{x} = \mathbf{b}$, $\mathbf{b} = (-3, 1, -15)$: replace column $i$ by $\mathbf{b}$, then $x_i = \det A_i / \det A$ (valid since $\det A = -1 \neq 0$).

$$A_1 = \begin{bmatrix} -3 & 2 & 3 \\ 1 & 5 & 3 \\ -15 & 0 & 8 \end{bmatrix} \xrightarrow{\text{expand col 1}} (-3)(40) - (1)(16) + (-15)(-9) = -120 - 16 + 135 = -1$$

$$A_2 = \begin{bmatrix} 1 & -3 & 3 \\ 2 & 1 & 3 \\ 1 & -15 & 8 \end{bmatrix} \xrightarrow{\text{expand col 2}} (-3)(-13) + (1)(5) + (-15)(3) = 39 + 5 - 45 = -1$$

$$A_3 = \begin{bmatrix} 1 & 2 & -3 \\ 2 & 5 & 1 \\ 1 & 0 & -15 \end{bmatrix} \xrightarrow{\text{expand col 2}} 2(31) + 5(-12) + 0 = 2$$

$$\mathbf{x} = \left( \frac{-1}{-1}, \frac{-1}{-1}, \frac{2}{-1} \right) = (1,\ 1,\ -2) \qquad \text{— exactly elimination's answer (slide 4) } \checkmark$$

- Cofactors recycled: 40, $-13$, 5, ... are the same numbers as in the adjugate on the next slide — one table of cofactors serves both.

# Ch. 3 step by step: all nine cofactors, the adjugate, $A^{-1}$

Minors $M_{ij}$ (delete row $i$, column $j$), cofactors $C_{ij} = (-1)^{i+j} M_{ij}$:

$$[M_{ij}] = \begin{bmatrix} 40 & 13 & -5 \\ 16 & 5 & -2 \\ -9 & -3 & 1 \end{bmatrix} \xrightarrow{\text{apply signs}} [C_{ij}] = \begin{bmatrix} 40 & -13 & -5 \\ -16 & 5 & 2 \\ -9 & 3 & 1 \end{bmatrix}$$

$$\xrightarrow{\text{transpose}} \operatorname{adj}(A) = \begin{bmatrix} 40 & -16 & -9 \\ -13 & 5 & 3 \\ -5 & 2 & 1 \end{bmatrix}$$

Then $A^{-1} = \operatorname{adj}(A)/\det A = \operatorname{adj}(A)/(-1)$:

$$A^{-1} = \begin{bmatrix} -40 & 16 & 9 \\ 13 & -5 & -3 \\ 5 & -2 & -1 \end{bmatrix}$$

identical to the row-reduction answer (slide 10) ✓

## Check the theorem $\operatorname{adj}(A)\, A = (\det A)\, I = -I$

Entry $(1,1)$: $40(1) + (-16)(2) + (-9)(1) = 40 - 32 - 9 = -1$ ✓ Entry $(1,2)$: $40(2) + (-16)(5) + (-9)(0) = 80 - 80 = 0$ ✓— that theorem is why $\det A$ divides out.

# The grand table: $A$ vs. $A'$

| Statement | $A$ | $A'$ |
|---|---|---|
| det (slides 15–16) | $-1$ | $0$ |
| pivots / rank | $3$ / rank $= 3$ | $2$ / rank $= 2$ |
| $A\mathbf{x} = \mathbf{b}$ (generic $\mathbf{b}$) | unique solution | none or infinitely many |
| $A\mathbf{x} = \mathbf{0}$ | only trivial | $z(-9, 3, 1) \neq \mathbf{0}$ |
| columns | independent, span $\mathbb{R}^3$ | dependent, span a plane |
| $T$: onto / one-to-one | yes / yes | no / no |
| inverse | row reduction and $\mathrm{adj}(A)/\det A$ agree | none |
| rank + dim Nul | $3 + 0 = 3$ | $2 + 1 = 3$ |
| volume $\lvert\det\rvert$ | $1$ | $0$ (flat) |

- One matrix and its twin: every midterm theorem is a row of this table, and every number in it was computed in this deck. Reproduce the table — and say *which theorem* each row invokes — and you are ready.

# Thank you!

Questions? Comments? Corrections?

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